

► IMMUNOLOGY—LYMPHOID STRUCTURES

Immune system organs

1° organs:

- Bone marrow—immune cell production, B cell maturation
- Thymus—T cell maturation

2° organs:

- Spleen, lymph nodes, tonsils, adenoids, appendix, Peyer patches
- Allow immune cells to interact with antigen

Lymph node

A 2° lymphoid organ that has many afferents, 1 or more efferents. Encapsulated, with trabeculae **A**. Functions are nonspecific filtration by macrophages, circulation of B and T cells, and immune response activation.

Follicle

Located in outer cortex; site of B-cell localization and proliferation. 1° follicles are dense and quiescent. 2° follicles have pale central germinal centers and are active.

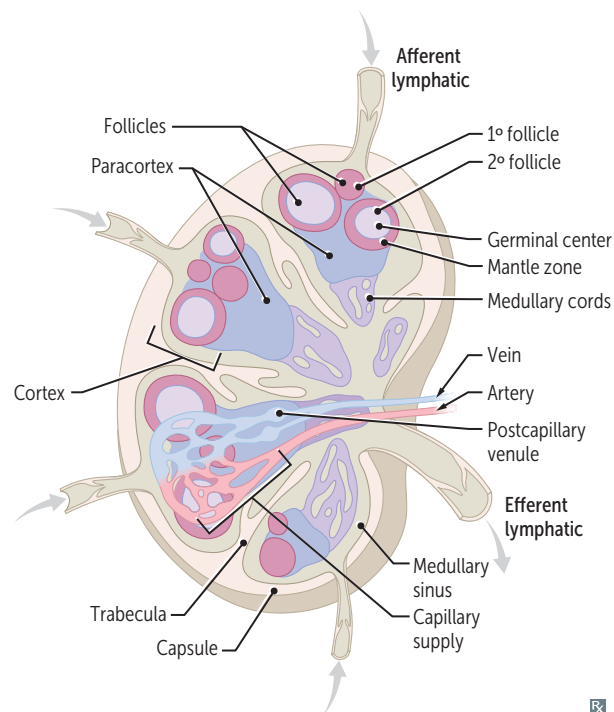
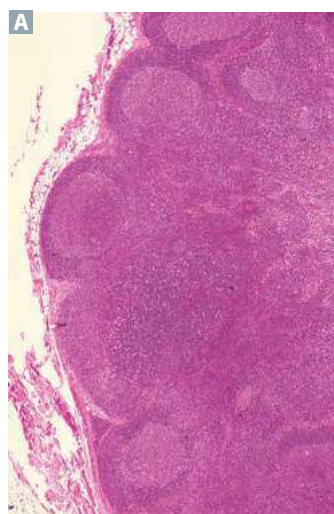
Medulla

Consists of medullary cords (closely packed lymphocytes and plasma cells) and medullary sinuses (contain reticular cells and macrophages). Medullary sinuses communicate with efferent lymphatics.

Paracortex

Contains T cells. Region of cortex between follicles and medulla. Contains high endothelial venules through which T and B cells enter from blood. Underdeveloped in patients with DiGeorge syndrome.

Paracortex enlarges in an extreme cellular immune response (eg, EBV and other viral infections → paracortical hyperplasia → lymphadenopathy).



Lymphatic drainage associations

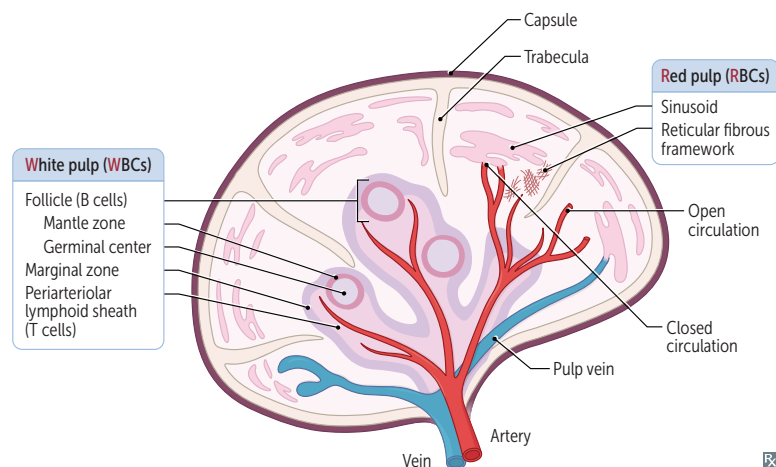
● Palpable lymph node
○ Nonpalpable lymph node

Lymph node cluster	Area of body drained	Associated pathology
Submandibular, submental	Oral cavity, anterior tongue, lower lip	Malignancy of and metastasis to the oral cavity
Deep cervical	Head, neck, oropharynx	Upper respiratory tract infection Infectious mononucleosis Kawasaki disease Malignancy of head, neck, oropharynx
Supraclavicular	Right: right hemithorax Left (Virchow node): left hemithorax, abdomen, pelvis	Malignancies of thorax, abdomen, pelvis
Mediastinal	Trachea, esophagus	Pulmonary TB (unilateral hilar) Sarcoidosis (bilateral hilar) Lung cancer Granulomatous disease
Hilar	Lungs	
Axillary	Upper limb, breast, skin above umbilicus	Mastitis Metastasis (especially breast cancer)
Epitrochlear	Hand, forearm	Secondary syphilis
Celiac	Liver, stomach, spleen, pancreas, upper duodenum	Mesenteric lymphadenitis Inflammatory bowel disease Celiac disease
Superior mesenteric	Lower duodenum, jejunum, ileum, colon to splenic flexure	
Inferior mesenteric	Colon from splenic flexure to upper rectum	
Periumbilical (Sister Mary Joseph node)	Abdomen, pelvis	Gastric cancer
Para-aortic	Pair of testes, ovaries, kidneys, fallopian tubes, fundus of uterus	Metastasis
External iliac	Body of uterus, cervix, superior bladder	Sexually transmitted infections Medial foot/leg cellulitis (superficial inguinal)
Internal iliac	Cervix, proximal vagina, corpus cavernosum, prostate, inferior bladder, lower rectum to anal canal (above pectinate line)	
Superficial inguinal	Distal vagina, vulva, scrotum, urethra, anal canal (below pectinate line), skin below umbilicus (except popliteal area)	
Popliteal ("pop-lateral")	Dorsolateral foot, posterior calf	Lateral foot/leg cellulitis

■ Right lymphatic duct drains right side of body above diaphragm into junction of the right subclavian and internal jugular vein

■ Thoracic duct drains below the diaphragm and left thorax and upper limb into junction of left subclavian and internal jugular veins (rupture of thoracic duct can cause chylothorax)

Spleen



Located in LUQ of abdomen, anterolateral to left kidney, protected by 9th-11th ribs. Splenic dysfunction (eg, postsplenectomy, sickle cell disease autosplenectomy) → ↓ IgM → ↓ complement activation → ↓ C3b opsonization → ↑ susceptibility to encapsulated organisms, against which patients should be vaccinated (from most to least common: pneumococci, meningococci, *Haemophilus influenzae* type b [Hib]).

Postsplenectomy findings:

- Howell-Jolly bodies (nuclear remnants)
- Target cells
- Thrombocytosis (loss of sequestration and removal)
- Lymphocytosis (loss of sequestration)

Periarteriolar lymphatic sheath

Contains T cells. Located within white pulp.

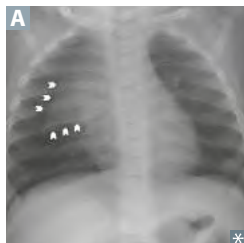
Follicle

Contains B cells. Located within white pulp.

Marginal zone

Contains macrophages and specialized B cells. Site where antigen-presenting cells (APCs) capture blood-borne antigens for recognition by lymphocytes. Located between red pulp and white pulp.

Thymus



Located in the anterosuperior mediastinum.

Site of T-cell differentiation and maturation.

Encapsulated. **T**hymus epithelium is derived from **t**hird pharyngeal pouch (endoderm), whereas thymic lymphocytes are of mesodermal origin. Cortex is dense with immature T cells; **m**edulla is pale with **m**ature T cells and Hassall corpuscles containing epithelial reticular cells.

Normal neonatal thymus “sail-shaped” on CXR (arrows in **A**), involutes by age 3 years.

T cells = **T**hymus

B cells = **B**one marrow

Absent thymic shadow or hypoplastic thymus seen in some immunodeficiencies (eg, SCID, DiGeorge syndrome).

Thymoma—neoplasm of **t**hymus. Associated with myasthenia gravis, superior vena cava syndrome, pure red cell aplasia, Good syndrome.